

Traffic Simulation Software Review

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Abstract— Metro Manila traffic, specifically in EDSA, needs to be simulated and analyzed in order to minimize waste and pollution. The purpose of this research is to compare different traffic simulation software and choose which is the most viable for the researchers to simulate traffic in EDSA. The software were compared based on the following criteria: (1) Open Source and Free Use, (2) Operating System Portability, (3) Creating traffic networks and associated vehicle patterns, (4) Quality of GUI and Documentation, (5) Simulation output (data and files), (6) Ability to simulate very large traffic networks, (7) Additional capabilities, and (8) CPU and Memory performance. SUMO, Vissim, and Aimsun were found to perform reasonably well and had common and unique features with different strengths and weaknesses.

I. INTRODUCTION

Traffic congestion in Metro Manila is a pressing issue the country has been facing for years. Based on a recent global evaluation conducted by Waze, the label “Worst Traffic on Earth” belonged to Metro Manila [9]. A number of traffic schemes have already been implemented by the government, having little to no effect on the congestion.

Due to notorious traffic jams in Metro Manila, a study showed that at least P3 billion pesos (\$64 million) a day is wasted on time consumed by people stuck in traffic and the extra cost of operating vehicles in gridlock [5].

Today’s advancement in technology gives people the ability to simulate environments at a rate much faster than any real environment. The result of the simulated environment provides results in minutes, hours, days or even weeks ahead of what the same experiment could provide if it were to be conducted in the real world.

One of the most commonly studied systems using a computer simulation is a traffic network since experimenting traffic on a real environment is not practical.

Computer simulations of traffic systems fall under the following two categories:

- 1) *Macroscopic simulation which includes statistical dispersion models, freeway traffic models, etc*
- 2) *Microscopic simulation which includes cellular automata, multi-agent simulation, particle system simulation, etc.*

In order to investigate different traffic strategies, four widely used and well-known simulation packages were reviewed:

1) *Matsim - Version 0.8.0*

Multi-Agent Transport Simulation is an open-source framework to implement large-scale agent-based transport simulations and transport planning models [8].

2) *SUMO - Version 0.28.0*

Simulation of Urban Mobility is an open-source, highly portable, microscopic and continuous road traffic simulation package design for large road networks [7].

3) *Aimsun - Version 8.1*

Aimsun is a traffic modelling software environment, which integrates travel demand modelling, macroscopic functionalities and the mesoscopic-microscopic hybrid simulator [4].

4) *PTV Vissim - Version 9*

Planung Transport Verkehr AG Verkehr In Städten - SIMulationsmodell is a microscopic multi-modal traffic flow simulation package that allows you to simulate exact traffic patterns and displays all road users and their interactions in one model [10].

II. STATE OF THE ART REVIEW

A comparison between AIMSUN and VISSIM was made by [11]. The study used both qualitative and quantitative evaluation criteria and incorporated room for user’s priorities

by assigning weights to each feature of evaluation [11]. AIMSUN models every single behavior continuously throughout the simulation period according to several driver behavior models such as car following, lane changing and gap-acceptance models. The model provides very detailed capabilities of modeling traffic network: it can distinguish between different types of vehicles; it can model traffic demands based on traffic flows and turning proportions or origin-destination (OD) matrices and route selection models; it can deal with a wide range of roadway types in arterial and freeway systems; it can also model impact of incidents and Variable Message Signs (VMS). AIMSUN can replicate any kind of traffic detector to collect the basic measurements such as vehicle counts, occupancy, presence, speed and density at any aggregation level defined by users. On the other hand, VISSIM is a time-step and behavior-based simulation model developed to model urban traffic and public transit operations. The model consists internally of two components that communicate through an interface. The first one is a traffic simulator, which is a microscopic traffic flow model that simulates the movement of vehicles and generates the corresponding output. The second components, named as the signal state generator, updates the signal status for the following simulation step. The model is capable of producing measures of effectiveness commonly used in the traffic engineering profession like total delay, stopped-time delay, stops, queue lengths, fuel emissions, and fuel consumption. The model has been successfully applied as a useful tool in a variety of transportation projects such as development and evaluation of transit signal priority logic, the evaluation and optimizations of traffic operations in a combined network of a coordinated and actuated traffic signals, etc. It was found that both simulators are capable of incorporating most of the standard features used in traffic modeling. The accuracy of both simulators was found to be similar. It was also observed that during the study, data entry for large networks is a time-consuming process. Several calibration methodologies exist for different models. Choosing between the two on what simulation model to use is upon the discretion of future researchers.

A review of traffic simulation software of [6] was published. Six simulation packages namely SUMO, Quadstone Paramics Modeller, Treiber's Microsimulation of Road Traffic, AIMSUN, Trafficware SimTraffic, and CORSIM TRAFVU, were studied and compared. The software mentioned were either open-source software or proprietary software. The software were compared based on the following criteria:

- 1) *Open Source and Free Use*
- 2) *Operating System Portability*
- 3) *Creating traffic networks and associated vehicle patterns*
- 4) *Quality of GUI & Documentation*
- 5) *Simulation output (data and files)*
- 6) *Ability to simulate very large traffic networks*
- 7) *Additional capabilities*
- 8) *CPU & Memory performance.*

These criteria were followed as the researchers conducted their methodology.

III. OPEN SOURCE & FREE USE

Open source software is software with source code that anyone can inspect, modify, and enhance [3]. This is why open source softwares are beneficial not just to programmers but to non-programmers too.

Two open source software, MATSim and SUMO and two proprietary software, PTV Vissim and Aimsun were selected for this study for better comparison. For the proprietary software, only the 30-day demo versions were utilized with limited features. To continue using Vissim, a student license was used that will last for the duration of this study. Meanwhile, MATSim and SUMO are freely available online.

MATSim's Java source code is available upon download and it has modules that can be replaced by own implementations to test different aspects of your own simulation. It also runs on all major operating systems like Windows, Unix/Linux, and etc. It also gives you the capability to extend the software using your own algorithms.

SUMO is implemented in C++ and only uses portable libraries [2]. Similar to MATSim, SUMO's source code is also available online. SUMO has a runnable application and a source code file that can be downloaded separately.

The advantage of MATSim and SUMO is that, programmers can modify them based on its usage and further improvements can also be suggested.

IV. OPERATING SYSTEM PORTABILITY

Portability is a characteristic attributed to a computer program if it can be used in an operating systems other than the one in which it was created without requiring major rework [1].

Among the simulation packages reviewed, 3 out of 4 can work on multiple operating systems. Sumo and MatSim can work on both Linux and Windows while Aimsun can work on Linux, Windows, and macOS. On the other hand, Vissim only works on Windows.

Operating System Portability is very important because it brings more users and gives existing users more options.

V. CREATING TRAFFIC NETWORKS AND ASSOCIATION VEHICLE PATTERNS

The four simulation software used have different approaches on creating traffic networks and vehicle patterns. The following are the different techniques under the software packages mentioned:

- SUMO
 - Manual write up of a traffic network in an XML file
 - Import networks from other simulation packages
 - Automatic network generator
- Vissim and Aimsun
 - Manually drawing of the traffic network using their graphical network editor

Each simulation package also has various vehicle pattern techniques.

- SUMO
 - Manual vehicle route definition which can be flow definitions, flow and turning ratios or Origin-Destination (OD) matrices
- Aimsun
 - OD Matrices using supply and demand of traffic
 - Vehicle route randomization
- Vissim
 - Vehicle routing decision
 - OD Matrices using supply and demand of traffic

VI. QUALITY OF GUI & DOCUMENTATION

The usability of a complex software such as a traffic simulation package relies on a well-explained documentation and an easy to understand user interface.

The four simulation packages offer documentation. MatSim, Vissim, and Aimsun have PDF documentations that can be downloaded online while MatSim has online documentation that is available on their website.

In regards to their Graphic User Interface (GUI), Aimsun and SUMO were easily used and understood. Vissim's GUI is comprehensive but it was manageable because of its very detailed documentation. On the other hand, MatSim was not utilized well because of the continuous errors during the package installation.

It needs to be noted that the GUI was rated based on a new user's perspective. Moreover, it is safe to assume that the existing users of the respective packages are already well-adapted to the environment of the software.

One of the best way for a non-technical user to see results of a traffic simulation is to actually view the simulation running in real time [6]. The graphical representation of a traffic network is very important to examine what really happens in a real world scenario. Through graphical representation, the user may be able to identify whether the currently implemented traffic schemes are effective or not.

Traffic simulation package displays are usually two dimensional, and among the traffic simulation packages mentioned only Aimsun and Vissim had three dimensional capabilities. Between Aimsun and Vissim, Vissim had the best graphical representation. It has more realistic vehicles, pedestrians, and even static city buildings, as pictured in Figure 1. The user can have a 360 view of the network while the simulation is running which helps in clearly examining the pattern of traffic flow.

Aimsun has a basic three dimensional preview of a traffic network. It also has a 360 view of the simulation, as shown in Figure 2, but the city buildings, cars, and pedestrians are not that detailed compared to that of Vissim.



Fig. 1. Vissim 3D



Fig. 2. Aimsun 3D

VII. SIMULATION OUTPUT

Statistical output is very important because it gives information that cannot be fully observed by just looking at graphical representations of a traffic simulation. That is why most of the simulation packages include some output files or data. The following includes the data and files found in each simulation package.

- SUMO
- Route Detectors: detector for sampling route distributions
 - Edge lane traffic: edge or lane-based network performance measures
 - Trip information: aggregated information about each vehicle's journey
 - Vehicle routes information: information about each vehicle's routes over simulation run
 - Simulation state statistics: information about the current state of the simulation
- Aimsun
 - Statistical simulation results: Network, Section, Node
 - Output database definition
 - Calculation of traffic statistics

- Experiment output
- Map output
- Vissim
 - Scenario Management: Comparing scenario results conveniently
 - Enhanced evaluation possibilities: Level of Service results for nodes
 - Improved handling of the matrices through matrix editor
 - Visualisation highlights
 - Vehicle record
 - Vehicle travel times (raw data)
 - Pedestrian record
 - Nodes (raw data)
 - Signal changes

VIII. ABILITY TO SIMULATE VERY LARGE TRAFFIC NETWORKS

Traffic networks are simulated to plan specific changes and observe its effects to a network. It may be a single road leading to a city or a very complex downtown area of a large city. That is why it is important for a simulation package to have an option of simulating very large traffic networks containing hundreds of roads and thousands of vehicles running.

SUMO, Aimsun, and Vissim has the ability to cope with large traffic networks. But as expected the larger the network the slower the frame rate was during simulation.

TABLE I. CPU & MEMORY USAGE

	CPU	RAM
Matsim	N/A	N/A
SUMO	30-40% (EDSA Kamuning Network)	12-16MB for average networks 1GB – EDSA Kamuning Network
Vissim	50-60% - 2D 65-70% - 3D (EDSA Balintawak Network)	720MB - 2D 780-800MB - 3D (EDSABalintawak Network)
Aimsun	30-40%, depending on the number of vehicles and the scenario currently simulated 15-20% 3D	300-400 MB - 2D, depending on the traffic network 1GB - 3D

IX. CONCLUSION

SUMO, Aimsun and Vissim can all provide reasonable results. All three simulation packages were found to perform reasonably well, but with limitations that should be understood before selecting one for network evaluation.

Vissim's ability to cope with large traffic networks makes it a good simulation package for simulating EDSA. The current traffic data provided by the Department of Transportation (DOTr), fits perfectly to the vehicle pattern techniques of Vissim.

In the end, there was no ranking of 'best' or 'worst' software. All three had common and unique features and different strengths and weaknesses. Knowing the data and the needs of the traffic network to be simulated is the key turning point to selecting a simulation package.

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